Ordered, disordered and hexatic phases of Graphene

Alexis ESCARMELLE

TP IV

January 2026

Professor: Oleg YAZYEV

School of Basic Sciences
Chair of Computational Condensed Matter Physics
Swiss Federal Institute of Technology in Lausanne

Contents

1 Introduction

Two-dimensional materials exhibit a rich variety of structural and orientational behaviors that have no direct analogue in three-dimensional solids. Because thermal fluctuations are particularly strong in 2D, crystalline order is fragile and may be lost through the proliferation of topological defects. In this context, the possibility of an intermediate hexatic phase, characterized by short-range translational order but quasi-long-range bond-orientational order—has attracted significant interest. While such a phase is central to the Kosterlitz-Thouless-Halperin-Nelson-Young (KTHNY) theory, its existence in realistic atomic 2D systems remains an open and actively discussed question.

Recent works on polycrystalline graphene growth, grain boundary energetics, and more recently on AgI embedded in multilayer graphene, have highlighted the role of dislocations, grain misorientations, and local orientational disorder. These studies provide motivation to explore simplified models in which grain orientation fields and defect-mediated interactions can be analyzed numerically.

The aim of this project is to investigate whether a hexatic-like regime can emerge in a minimal model inspired by the Hexatic Graphene draft. The work follows three main steps. First, we reviewed the relevant literature to identify the essential mechanisms governing orientational order and the role of misorientation between grains. Second, we implemented a Monte Carlo simulation tailored to this problem and validated the numerical approach using the classical XY model. Finally, we applied the algorithm to our polycrystalline system in order to compute the local bond-orientational order parameter and its correlation function , and to determine under which conditions the system may exhibit signatures of hexatic behavior.

This report summarises the methodology used to construct and simulate the system, the observables chosen to characterize orientational order, and the results obtained for different grain configurations and system sizes. The broader goal is not to provide a full physical model of 2D melting, but rather to assess how far a simplified Monte Carlo framework can reproduce features reminiscent of a hexatic phase.

2 Polycristalline Model

2.1 Hamiltonian

In this project, we focus on a two-dimensional system of grains (crystallites) arranged in a hexagonal lattice structure. Each grain has an orientation angle, denoted by θ_i, which represents its deviation from the substrate's preferred orientation. This configuration corresponds to a graphene sheet placed on a hexagonal substrate.

In this system, the distribution of grain orientations is governed by two competing interactions: the substrate coupling [1], which tends to align the grains with the substrate, and the grain boundary energy, which favors local alignment between neighboring grains. This interaction is expressed according to Read-Shockley model [2]. The total Hamiltonian $\mathcal{H}$ of the system can be written as:

$$\mathcal{H} = \frac{\epsilon}{\rho} \sum_i \theta_i^2 + \frac{\gamma}{3\rho} \sum_i \sum_{j \in \mathrm{nn}_i} |\Delta\theta_{ij}| \left(A - \log |\Delta\theta_{ij}| \right) \tag{1}$$

Where ϵ is the substrate coupling strength, γ is the grain boundary energy coefficient, ρ is the grain density and A is a constant related to the material properties. The term $\Delta\theta_{ij} = \theta_i - \theta_j$ represents the difference in orientation angles between neighboring grains i and j, with the summation over nn_i indicating that we only consider nearest neighbors.

2.2 Observables

We define several observables to characterize the phases in the system as described in the KTHNY theory [3–6].

To characterize the orientational order in the system, we define the local bond-orientational order parameter $\psi_6(\mathbf{r}_i)$ for each grain i as:

$$\psi_6(\mathbf{r}_i) = e^{i6\theta_i} \tag{2}$$

The bond-orientational correlation function $G_6(r)$ between grains separated by a distance r is defined as:

$$G_6(r) = \langle \psi_6(\mathbf{r}_i) \psi_6^*(\mathbf{r}_j) \rangle_{|\mathbf{r}_i - \mathbf{r}_j| = r} \tag{3}$$

Where the average is taken over all pairs of grains (i, j) separated by distance r. The behavior of $G_6(r)$ provides information about the orientational order of the system. In a hexatic phase, $G_6(r)$ is expected to decay as a power law:

$$G_6(r) \sim r^{-\eta_6} \tag{4}$$

Where η_6 is a critical exponent characterizing the decay of orientational correlations. At the transition between hexatic and disordered phases, η_6 reaches a universal value of 1/4. In contrast, in a disordered phase, $G_6(r)$ decays exponentially:

$$G_6(r) \sim e^{-r/\xi_6} \tag{5}$$

Where ξ_6 is the orientational correlation length. In an ordered phase, $G_6(r)$ must be equal to a constant for large r, indicating long-range orientational order.

In the case of $\gamma = 0$, the correlation function $G_6(r)$ can be computed analytically as:

$$G_6(r) = \exp\left(-18\rho\frac{k_B T}{\epsilon}\right) \tag{6}$$

The susceptibility associated with the bond-orientational order parameter, χ_6, is defined as:

$$\chi_6 = N\beta\left(\langle|\Psi_6|^2\rangle - \langle|\Psi_6|\rangle^2\right) \tag{7}$$

Where N is the total number of grains in the system, $\beta = \frac{1}{k_B T}$ and $\Psi_6 = \frac{1}{N}\sum_i \psi_6(\mathbf{r}_i)$ is the global bond-orientational order parameter. This susceptibility measures the fluctuations in the bond-orientational order parameter and is expected to diverge at the phase transition between ordered and disordered phases.

3 Simulation Method

To study the behavior of the polycrystalline graphene system described by the Hamiltonian above, we employ a Monte Carlo simulation approach using the Metropolis algorithm. The simulation is performed on a two-dimensional hexagonal lattice of size $L \times L$, where each lattice site represents a grain with an associated orientation angle θ_i. In addition to the standard Metropolis updates, we implement over-relaxation moves to enhance the sampling efficiency and reduce autocorrelation times. A cluster algorithm is also used at low temperatures to pass energy barriers and improve convergence. These methods are performed every choosen n steps and are detailed for the XY model in the book *Statistical Mechanics: Algorithms and Computations* by Werner Krauth [7].

Algorithm 1 Standard Monte Carlo Step

1: Choose a random grain (i, j)
2: Propose a new orientation angle $\theta'_{ij} = \theta_{ij} + \delta\theta$ with $\delta\theta \in [-\Delta, \Delta]$
3: Compute the energy difference $\Delta E = \mathcal{H}(\theta'_{ij}) - \mathcal{H}(\theta_{ij})$
4: Accept the new angle with probability $P_{\text{accept}} = \min(1, e^{-\beta \Delta E})$

Algorithm 2 Over-Relaxation Step

1: **for** each grain i **do**
2: Compute the local effective field $h_i = \sum_{j \in \text{nn}_i} J_{ij}\theta_j$
3: Update the orientation angle $\theta'_i = 2\frac{h_i}{6} - \theta_i$
4: Compute the energy difference $\Delta E = \mathcal{H}(\theta'_i) - \mathcal{H}(\theta_i)$
5: Accept the new angle with probability $P_{\text{accept}} = \min(1, e^{-\beta \Delta E})$
6: **end for**

Algorithm 3 Cluster Update Step

1: Choose a random seed grain and initialize the cluster $\mathcal{C}$
2: Grow the cluster by adding neighboring grains with small angle mismatchif $|\Delta\theta_{ij}| < \text{tol}$, add grain j to cluster $\mathcal{C}$
3: Estimate the collective rotation angle $\Delta\theta$ (median of boundary mismatches)
4: Compute the energy change associated with rotating the whole cluster
5: Accept the rotation with probability $P_{\text{accept}} = \min(1, e^{-\beta \Delta E})$

Then, the simulation proceeds as follows:

Algorithm 4 Monte Carlo Simulation Procedure

Initialize the hexagonal lattice with random grain orientations $\theta_i \in [-\pi/6, \pi/6]$ with open boundary conditions
for n_{steps_therm} **do**
    Perform a standard Monte Carlo step
    Perform an over-relaxation step every $n_{overrelax}$ steps
    Perform a cluster update step every $n_{cluster}$ steps
    Adjust Δ to maintain an acceptance ratio of $\sim 50\%$ every n_{tune} steps
end for
for n_{steps_meas} **do**
    Perform a standard Monte Carlo step
    Measure and save observables every $n_{measure}$ steps
end for

4 Test on the XY model

The first step was to validate the Monte Carlo simulation code by applying it to the classical XY model, which is a well-studied system. The Hamiltonian of the XY model is given by:

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j) \tag{8}$$

5 $\epsilon = 1.0, \gamma = 0.0$

Once the simulation code was validated for the XY model, we proceeded to test it for the case where $\epsilon = 1.0$ and $\gamma = 0.0$. In this case, the system should not exhibit any hexatic order and the correlation function $G_6(r)$ is expected to be constant.

6 $\epsilon = 0.0, \gamma = 1.0$

References

[1] P. C. Rogge, K. Thürmer, M. E. Foster, et al., "Real-time observation of epitaxial graphene domain reorientation," *Nature Communications*, vol. 6, no. 1, p. 6880, 2015. DOI: 10.1038/ncomms7880 [Online]. Available: https://doi.org/10.1038/ncomms7880

[2] W. T. Read and W. Shockley, "Dislocation models of crystal grain boundaries," *Phys. Rev.*, vol. 78, pp. 275–289, 3 May 1950. DOI: 10.1103/PhysRev.78.275 [Online]. Available: https://link.aps.org/doi/10.1103/PhysRev.78.275

[3] B. I. Halperin and D. R. Nelson, "Theory of two-dimensional melting," *Phys. Rev. Lett.*, vol. 41, pp. 121–124, 2 Jul. 1978. DOI: 10.1103/PhysRevLett.41.121 [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevLett.41.121

[4] D. R. Nelson and B. I. Halperin, "Dislocation-mediated melting in two dimensions," *Phys. Rev. B*, vol. 19, pp. 2457–2484, 5 Mar. 1979. DOI: 10.1103/PhysRevB.19.2457 [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevB.19.2457

[5] J. M. Kosterlitz and D. J. Thouless, "Ordering, metastability and phase transitions in two-dimensional systems," *Journal of Physics C: Solid State Physics*, vol. 6, no. 7, p. 1181, Apr. 1973. DOI: 10.1088/0022-3719/6/7/010 [Online]. Available: https://doi.org/10.1088/0022-3719/6/7/010

[6] A. P. Young, "Melting and the vector coulomb gas in two dimensions," *Phys. Rev. B*, vol. 19, pp. 1855–1866, 4 Feb. 1979. DOI: 10.1103/PhysRevB.19.1855 [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevB.19.1855

[7] W. Krauth, *Statistical Mechanics: Algorithms and Computations* (EBSCO ebook academic collection). OUP Oxford, 2006, ISBN: 9780198515364. [Online]. Available: https://books.google.ch/books?id=g30SDAAAQBAJ